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Bus tracker app development

Design and Implementation of a Real-Time Bus Tracker Application: Technologies, Integration, and User Experience

Quick Reference

Key Findings Table

Theme	Key Insights	Supporting Citations
Technologies & Architectures	Multi-tier architectures with GPS, IoT sensors, cloud/edge computing, and advanced filtering for real-time tracking.	1 2 3 4 5
User Experience & Accessibility	User-centered design, accessibility for visually impaired, real-time notifications, and participatory design are critical.	6 7 8 9 10
Data Integration & Fusion	Heterogeneous data sources (GPS, RFID, IoT, smart cards) require advanced fusion (Kalman, Dempster–Shafer, ML).	11 12 13 14
Scalability & Cloud	Cloud/microservices architectures (AWS, Firebase), MQTT/LoRaWAN protocols, and edge computing for scale/latency.	15 16 17 18
Real-Time Feedback & Iteration	Continuous user feedback loops drive iterative improvements, increasing trust and reducing uncertainty.	19 20 21 22
Case Studies & Limitations	Real-world deployments show improved efficiency but face GPS noise, hardware faults, and harsh condition limits.	3 8 19 23 24

Direct Answer

For building a bus tracker app, consider reading:

- Studies on GPS and IoT integration for real-time tracking 1 2 3
- Research on cloud-based architectures and low-latency data processing 3 15 17
- User experience and accessibility-focused papers 6 10
- Case studies and performance evaluations 19 23

- Integration methodologies and standards for ticketing and monitoring [25] [26]

Study Scope

- **Time Period:** 2015–2024 (focus on recent advances)
- **Disciplines:** Computer Science, Transportation Engineering, Human-Computer Interaction, Urban Informatics
- **Methods:** Empirical studies, meta-analyses, case studies, prototyping, user-centered design, algorithmic benchmarking

Assumptions & Limitations

- Most studies focus on urban and university bus systems; rural deployments are less represented.
- GPS accuracy and communication reliability are assumed to be sufficient for most urban deployments, but harsh environments (dense urban, rural, or adverse weather) present ongoing challenges.
- User experience research often centers on sighted users; accessibility for visually impaired and marginalized groups is an emerging but underexplored area.
- Integration standards (e.g., GTFS, SIRI, MQTT) are recommended but not universally adopted, leading to interoperability gaps.

Suggested Further Research

- Standardized, secure, and scalable protocols for seamless integration of ticketing, tracking, and monitoring.
- Advanced sensor fusion (AI, blockchain, federated learning) for robust, privacy-preserving real-time tracking.
- Participatory and inclusive design methods for marginalized and disabled users.
- Longitudinal studies on the impact of real-time feedback and iterative app improvement on ridership and operational efficiency.

1. Introduction

Background and Motivation

The rapid urbanization of cities and the increasing demand for efficient public transportation have driven the development of real-time bus tracking applications. These systems aim to enhance operational efficiency, passenger safety, and user satisfaction by providing accurate, timely information on bus locations, estimated arrival times (ETAs), and service disruptions. The motivation for developing such applications stems from the need to reduce

commuter anxiety, optimize fleet management, and support inclusive mobility for all user groups, including those with disabilities [1] [6] [11].

2. Theoretical Frameworks

Core Technologies and Multi-Tier Architectures

Modern bus tracking systems are built on multi-tier architectures that integrate several technological layers:

- **Data Acquisition:** GPS modules, IoT sensors (accelerometers, gyroscopes, RFID, PIR, temperature, flame sensors) collect real-time data on bus location, passenger presence, and environmental conditions [1] [2] [3] [4].
- **Communication:** Wireless protocols such as GSM, WiFi, LoRaWAN, and MQTT transmit data from buses to central servers or cloud platforms [5] [27].
- **Processing:** Cloud servers and edge computing nodes process, filter, and analyze incoming data, supporting advanced features like ETA prediction and route optimization [28] [29].
- **User Interface:** Mobile and web applications provide real-time information, notifications, and interactive features to passengers and operators [15] [16].

This layered approach enables modularity, scalability, and efficient data flow, supporting both operational needs and user experience [15] [16].

Cloud-Based and Edge Computing Solutions

Cloud platforms (e.g., AWS, Firebase) and edge computing architectures are increasingly adopted to handle the scalability and latency requirements of large metropolitan bus tracking systems. Edge computing processes data closer to the source, reducing latency and bandwidth usage, while cloud platforms provide elastic storage and computational resources for analytics and historical data management [28] [29] [30] [31].

Advanced Algorithms for ETA Prediction and Route Optimization

Beyond traditional Kalman filters, machine learning models such as LSTM, Random Forest, SVM-Kalman hybrids, and ensemble methods (XGBoost, CatBoost) are used for ETA prediction and dynamic route optimization. These models leverage historical and real-time data to improve prediction accuracy and adapt to changing traffic conditions [32] [33] [34] [35].

IoT Sensor Fusion and Integration

Sensor fusion techniques, including extended Kalman filtering, transformer networks, and Dempster–Shafer theory, combine data from multiple sensors (GPS, IMU, RFID, BLE, WiFi) to enhance location accuracy and system reliability. These frameworks can achieve real-time positioning accuracy above 99% in controlled deployments [5] [36] [37] [38].

Communication Protocols and Power Efficiency

- **MQTT:** Lightweight, low-latency, suitable for real-time updates in both urban and rural settings.

- **LoRaWAN:** Low power, long-range, ideal for rural or campus environments with limited infrastructure [5] [27] [39].

Synthesis: The convergence of IoT, cloud/edge computing, and advanced algorithms forms the backbone of modern bus tracking systems, enabling real-time, scalable, and reliable service delivery [1] [2] [3] [4] [5].

3. Methods & Data Transparency

Data Sources and Fusion Methods

Bus tracking systems integrate heterogeneous data sources:

- **GPS:** Primary source for real-time location.
- **RFID:** Used for bus identification, stop detection, and automated fare collection.
- **IoT Sensors:** Monitor environmental and operational parameters (e.g., temperature, passenger count).
- **Smart Cards/QR Codes:** Capture passenger travel behavior and support ticketing [11] [12] [13] [14].

Data fusion frameworks (e.g., Kalman filters, Dempster–Shafer, adaptive kernel smoothing) are employed to reconcile noisy or incomplete data, improving accuracy and reliability [13] [14].

Integration of Ticketing, Tracking, and Monitoring

Integration challenges include:

- Lack of standardized protocols and operational procedures.
- Reliance on manual processes and disparate hardware/software platforms.
- Need for seamless, secure data connectivity across devices and cloud servers [26] [40] [41] [42].

Standardized protocols such as GTFS, SIRI, and MQTT are recommended for interoperability, but adoption varies [26] [41].

Microservices and Cloud Platforms

Microservices architectures (e.g., FastAPI, RESTful services) and cloud platforms (AWS, Firebase) support modular, scalable, and real-time data processing. These architectures facilitate independent deployment, rapid iteration, and integration of new features without disrupting core services [15] [16] [17] [18].

Security and Reliability in Data Transmission

Best practices include:

- Encryption (AES, HMAC), blockchain integration, and fault-tolerant communication strategies.
- Physical-layer security (e.g., cooperative jamming, fountain coding) for vehicular networks.

- Redundant routing and standby switching for fault tolerance [43] [44] [45].

Synthesis: Robust data integration and secure, scalable architectures are essential for reliable, real-time bus tracking, especially as systems grow in complexity and scale [11] [12] [13] [14] [17] [18].

4. Critical Analysis of Findings

User Experience and Accessibility in Bus Tracking Apps

Key User Experience Factors

- Real-time notifications, accurate ETAs, and user-friendly interfaces are critical for reducing commuter anxiety and improving satisfaction [6] [7] [8] [9].
- Safety features (e.g., emergency alerts, live monitoring) and additional information (driver details, route specifics) further enhance user trust [8] [46].

Accessibility for Visually Impaired Users

- Speech recognition, text-to-speech, and multimodal feedback (audio, haptic) are effective for visually impaired users [10].
- Participatory design and scenario-based prototyping involving visually impaired users lead to more accessible and user-centered solutions [47] [48].

Incorporating Diverse Passenger Personas

- Data-driven profiling (smart card data, cluster analysis) and persona-based toolkits help tailor app features to diverse user needs [49] [50] [51].
- AI-supported persona generation and iterative prototyping improve inclusivity and usability [19] [52].

Feedback Mechanisms and Anxiety Reduction

- Real-time delay notifications and user-friendly feedback mechanisms reduce travel anxiety and increase satisfaction [53] [54] [55] [56].
- Personalized, flexible feedback (visual, haptic, metaphorical) enhances engagement and emotional regulation [57] [58].

Synthesis: User-centered, accessible design—supported by real-time feedback and participatory methods—directly impacts the adoption and effectiveness of bus tracking apps, especially for marginalized groups [6] [10] [49] [53].

5. Real-world Implications

Case Studies and Performance Evaluations

Effectiveness and Limitations of Existing Systems

- User-centered apps (e.g., Transit Track, university bus systems) improve real-time visibility and commuter experience [3] [19].
- GPS data inaccuracies (multipath, NLOS, signal loss) and hardware faults remain significant challenges, especially in dense urban or harsh environments [8] [23] [59].
- Advanced filtering (Kalman, ML-based) and data preprocessing mitigate some errors but require ongoing refinement [60] [61] [62].

User Feedback and Iterative Improvement

- Structured surveys, prototype testing, and real-time feedback loops drive iterative improvements, increasing usability and satisfaction [19] [63].
- Multi-channel feedback (SMS, app-based) ensures reliability even in areas with poor connectivity [64].

IoT Sensor Integration for Operational Control and Safety

- IoT sensors (GPS, RFID, environmental, facial recognition) enable real-time monitoring, geo-fencing, and safety alerts, improving operational control and passenger security [5] [65] [66].

Addressing Limitations in Harsh Operating Conditions

- Alternative communication (LoRa), advanced filtering, and smartphone-based apps with cloud databases (Firebase) address limitations in connectivity, accuracy, and cost [8] [24] [67].

Synthesis: Real-world deployments validate the effectiveness of integrated, user-centered bus tracking systems but highlight persistent challenges in data accuracy, system reliability, and inclusivity—especially under adverse conditions [3] [8] [19] [23] [24].

6. Future Research Directions

Emerging Technologies and Integration Challenges

- **AI & Blockchain:** AI-driven sensor fusion, federated learning, and blockchain for secure, privacy-preserving data sharing are promising but require further research for robust, scalable deployment [5] [42] [44].
- **Standardization:** There is a critical need for standardized, interoperable protocols to harmonize ticketing, tracking, and monitoring across heterogeneous systems [26] [41] [42].

Enhancing Accessibility and Inclusivity

- Participatory design and adaptive interfaces for marginalized and disabled users remain underexplored. Research should focus on multimodal feedback, scenario-based prototyping, and inclusive persona development [47] [68].

User-Centered Innovation and Evaluation

- Continuous user involvement, iterative prototyping, and comprehensive evaluation are essential for driving innovation and ensuring that bus tracking apps meet evolving user needs [69] [70] [71].

Synthesis: Future research should prioritize secure, standardized integration, advanced sensor fusion, and inclusive, user-centered design to address current gaps and support the next generation of real-time bus tracking applications [5] [44] [47] [69].

7. Conclusion

Summary of Key Insights

- **Technological Foundations:** Multi-tier architectures, IoT sensor fusion, and cloud/edge computing underpin modern bus tracking systems, enabling real-time, scalable, and reliable service [1] [2] [3] [4].
- **Integration & Security:** Advanced data fusion, microservices, and standardized protocols are essential for integrating heterogeneous data sources and ensuring secure, reliable operation [11] [12] [13] [14].
- **User Experience:** User-centered, accessible design—supported by real-time feedback and participatory methods—directly impacts adoption, satisfaction, and inclusivity [6] [7] [8] [9].
- **Real-World Impact:** Case studies confirm operational and user experience benefits but highlight ongoing challenges in GPS accuracy, system reliability, and inclusivity, especially in harsh environments [3] [8] [19] [23].
- **Research Gaps:** Standardized integration, advanced sensor fusion, and inclusive design for marginalized users are key areas for future research [5] [44] [47].

Final Synthesis: The interplay between advanced technology, robust integration, and user-centered design is central to building effective, scalable, and inclusive bus tracking applications. Addressing current gaps in standardization, sensor fusion, and accessibility will be critical for the next generation of smart urban mobility solutions [1] [6] [11] [19].

References

1. Vehicle Forecasting: A Machine Learning Approach for Predicting and Preventing Vehicle Issues, Based on the Design Thinking Approach Priyadharshini, S., Revathi, K., Sowndharya, S., (...), Thayalini, K. Proceedings - 2024 International Conference on Expert Clouds and Applications, ICOECA 2024, 2024 https://www.scopus.com/pages/publications/85202342040?origin=scopusAI

2. Enhancing Emergency Response in Transit Using Cloud-Connected Bus Tracking for Safety and Medical Assistance Raman, R., Gurpur, S. 2024 2nd International Conference on Disruptive Technologies, ICDT 2024, 2024 https://www.scopus.com/pages/publications/85191252888?origin=scopusAI

3. Real-time Full-Stack University Bus Tracking System Basedon IoT Pabel, M.D., Rahman, M.S., Parvin, M.N., (...), Rahman, M. Intelligent Networks and Systems: Advanced Technologies and Applications, 2025 https://www.scopus.com/pages/publications/105021605311?origin=scopusAI

4. Smart Bus Real-Time Tracking System Using GSM and GPS Module Mundada, K., Patti, S., Rajguru, T., (...), Shambharkar, S. Lecture Notes in Networks and Systems, 2023 https://www.scopus.com/pages/publications/85174448752?origin=scopusAI

5. GPS-Driven IoT based Bus Monitoring System to ensure Passenger Security Nair, R.R., Srinivasan, H., Ramani, J., (...), Lavanyasri, A. 2025 7th International Symposium on Advanced Electrical and Communication Technologies, ISAECT 2025, 2025 https://www.scopus.com/pages/publications/105032697957?origin=scopusAI

6. Communicating User Insights with Travel Mindsets and Experience Personas in Intra-city Bus Context Hildén, E., Väänänen, K. Lecture Notes in Computer Science, 2019 https://www.scopus.com/pages/publications/85072957290?origin=scopusAI

7. A bus tracking and planning system for ctfastrak Wu, S., Carroll, S.M., Boyd-Carter, K., Krostoski, J. ASEE Annual Conference and Exposition, Conference Proceedings, 2014 https://www.scopus.com/pages/publications/84905193158?origin=scopusAI

8. Real-Time College Bus Tracking Application Designed for Hybrid Smartphones Himanshu, Singh, P., Hussain, M.A., (...), Krishnan, T.S.H. 2025 International Conference on Intelligent Control, Computing and Communications, IC3 2025, 2025 https://www.scopus.com/pages/publications/105003904023?origin=scopusAI

9. Design and Development of Bus Tracker: A Mobile App for Tracking Buses in Real Time Yusof, S.A.M., Kuan, L.P. Proceedings of 6th International Conference on Internet Applications, Protocols and Services: Enduring Internet's Use and Growth in the Pandemic Era, NETAPPS 2020, 2021 https://www.scopus.com/pages/publications/85180130334?origin=scopusAI

10. Bus search: Development and evaluation of performance and usability of mobile application for blind or visually impaired Viyanon, W. Mendel, 2014 https://www.scopus.com/pages/publications/84938085364?origin=scopusAI

11. Advance Public Bus Transport Management System: An Innovative Smart Bus Concept Patel, K., Patel, D., Chaudhary, N., (...), Mistry, A. Digest of Technical Papers - IEEE International Conference on Consumer Electronics, 2024 https://www.scopus.com/pages/publications/85186970607?origin=scopusAI

12. Road Traffic Density Estimation Based on Heterogeneous Data Fusion Zisner, P., Rettore, P.H.L., Santos, B.P., (...), Sevenich, P. Proceedings - IEEE Symposium on Computers and Communications, 2022 https://www.scopus.com/pages/publications/85141179535?origin=scopusAI

13. Performance analysis of data fusion methods for radar and IRST 3D target tracking Rajeswari, K., Ishwarya, A., Vaishnavi, K.K., Thiruvengadam, S.J. Proceedings of the 2017 International Conference on Wireless Communications, Signal Processing and Networking, WiSPNET 2017, 2017 https://www.scopus.com/pages/publications/85046424301?origin=scopusAI

14. A Review of Multi-sensor Data Fusion for Traffic Zhao, X., Zhang, D. Communications in Computer and Information Science, 2018 https://www.scopus.com/pages/publications/85051135932?origin=scopusAI

15. Real-Time campus university bus tracking mobile application Sharif, S.A., Suhaimi, M.S., Jamal, N.N., (...), Jawawi, D.N.A. Proceeding of 2018 7th ICT International Student Project Conference, ICT-ISPC 2018, 2018 https://www.scopus.com/pages/publications/85058190466?origin=scopusAI

16. Geo-Fencing based Serverless Smart Bus Notification System Mohana Priya, S., Subha, J. Proceedings of 4th International Conference on Electronics and Renewable Systems, ICEARS 2026, 2026 https://www.scopus.com/pages/publications/105035730254?origin=scopusAI

17. Interdependent Microservice Offloading and Semantic Aware Results Transmission for 6G Vehicular Edge Networks Salah ud din, M., Ali, M.N., Raza, G.M., (...), Kim, B.-S. IEEE Transactions on Intelligent

Transportation Systems, 2026 https://www.scopus.com/pages/publications/105038796859?origin=scopusAI

18. A microservices approach to scenario-based integration of smart mobility digital twins Nicolas, D., Marques, R.J., Feltus, C. Proceedings of SPIE - The International Society for Optical Engineering, 2025 https://www.scopus.com/pages/publications/105026445251?origin=scopusAI

19. Transit Track: A Mobile Application for Smarter Public Bus Transportation using Design Thinking Approach A User-Centered Approach to Modernizing Public Transportation in the Philippines Intal, G.L., Cubar, V.H., Reyes, D.J., (...), Tan, M.E. ICSIM 2026 - Proceedings of 2026 the 9th International Conference on Software Engineering and Information Management, 2026 https://www.scopus.com/pages/publications/105039448380?origin=scopusAI

20. Cooperative transit tracking using smart-phones Thiagarajan, A., Biagioni, J., Gerlich, T., Eriksson, J. SenSys 2010 - Proceedings of the 8th ACM Conference on Embedded Networked Sensor Systems, 2010 https://www.scopus.com/pages/publications/78650869210?origin=scopusAI

21. Usability Evaluation - User Experience Indonesia Train Ticket Reservation: System Usability Scale (SUS) and Think-Aloud Protocol Jati, L.A., Azzahra, A.P., Putra, E.P., Radhia, F. 2025 5th International Conference on Intelligent Cybernetics Technology and Applications, ICICyTA 2025, 2025 https://www.scopus.com/pages/publications/105033635573?origin=scopusAI

22. The development of a mobile map application for park and ride users Tanaksaranond, G. International Journal of GEOMATE, 2018 https://www.scopus.com/pages/publications/85048787335?origin=scopusAI

23. Enhancing Bus Performance Analysis Using GPS Data: Insights and Recommendations Tendulkar, S.G. International Conference on Intelligent Communication Networks and Computational Techniques, ICICNCT 2025, 2025 https://www.scopus.com/pages/publications/105030466428?origin=scopusAI

24. Leveraging LSTM for Accurate ETA Classification and Delay Prediction in College Bus Tracking Systems Lakshmi, R.A., Kaviyashri, V., Sri, M.L., (...), Rubika, R. Proceedings - 3rd International Conference on Self Sustainable Artificial Intelligence Systems, ICSSAS 2025, 2025 https://www.scopus.com/pages/publications/105012823383?origin=scopusAI

25. Efficient Smart Public Transit System Using RFID and IoT integration with MERN web applications Madasanal, K., Gudnavar, A., Nagarmunoli, S., (...), Naregal, K. ICCECE 2025 - International Conference on Computer, Electrical and Communication Engineering, 2025 https://www.scopus.com/pages/publications/105002922815?origin=scopusAI

26. Analysis of Traditional and Promising Methods for Managing Passenger Mobility in Urban Bus Systems Jumabaev, D., Allanazarov, U., Abdikhalilov, F., (...), Arndt, W.-H. Proceedings of International Conference on Applied Innovation in IT, 2026 https://www.scopus.com/pages/publications/105051888123?origin=scopusAI

27. A comprehensive review of energy harvesting and routing strategies for IoT sensors sustainability and communication technology Aldin, H.N.S., Ghods, M.R., Nayebipour, F., Torshiz, M.N. Sensors International, 2024 https://www.scopus.com/pages/publications/85178493459?origin=scopusAI

28. A Real-Time Bus Positioning System Based on LoRa Technology Guan, P., Zhang, Z., Wei, L., Zhao, Y. 2nd International Conference on Smart Grid and Smart Cities, ICSGSC 2018, 2018 https://www.scopus.com/pages/publications/85059936033?origin=scopusAI

29. A Real-Time Machine Learning-Based Public Transport Bus-Passenger Information System Skhosana, M., Ezugwu, A.E. International Journal of Sustainable Development and Planning, 2021 https://www.scopus.com/pages/publications/85121766197?origin=scopusAI

30. Intelligent Edge Computing Architecture: Low-Latency Transmission in an Intelligent Transport System for IoT Applications Tamariz-Flores, E.I., Torrealba-Meléndez, R., Muñoz-Pacheco, J.M., (...), Arriaga-Arriaga, C.A. IoT,

2026 https://www.scopus.com/pages/publications/105042718272?origin=scopusAI

31. Coral-pie: A geo-distributed edge-compute solution for space-time vehicle tracking Xu, Z., Shah, H.S., Ramachandran, U. Middleware 2020 - Proceedings of the 2020 21st International Middleware Conference, 2020 https://www.scopus.com/pages/publications/85098524920?origin=scopusAI

32. Hybrid model for prediction of bus arrival times at next station Yu, B., Yang, Z.-Z., Chen, K., Yu, B. Journal of Advanced Transportation, 2010 https://www.scopus.com/pages/publications/77958074007?origin=scopusAI

33. Deep Learning-Based Approach for Public Transportation Estimated Time Arrival Prediction Munkhbayar, T., Dashdorj, Z., Demchigjav, B., (...), Altangerel, E. Proceedings of the 2025 International Conference on Advanced Machine Learning and Data Science, AMLDS 2025, 2025 https://www.scopus.com/pages/publications/105019056156?origin=scopusAI

34. INTERPRETABLE MACHINE LEARNING FOR IDENTIFYING THE DETERMINANTS OF BUS DELAYS MARTYANOV, V., BUDZYńSKI, A., CZEREPICKI, A., KOZŁOWSKI, M. Archives of Transport, 2026 https://www.scopus.com/pages/publications/105047243133?origin=scopusAI

35. Machine Learning-Based Framework for Predicting Vehicle Arrival Times in Public Transport Ramanayake, L., Ranasinghe, H., Jayasinghe, S., (...), Karunatilake, C. ICAC 2025 - 7th International Conference on Advancements in Computing: The Future of Computing; AI, Quantum, and Beyond, 2025 https://www.scopus.com/pages/publications/105033581647?origin=scopusAI

36. Wireless-Assisted Automatic Online Spatial Calibration based on 5G TSN for Sensor Fusion Lei, M., Rahman, M., Mathaikutty, D., Alvarez, I. Proceedings of 2023 IEEE International Conference on Signal Processing, Communications and Computing, ICSPCC 2023, 2023 https://www.scopus.com/pages/publications/85184851806?origin=scopusAI

37. Multi-Source Sensor Fusion for Robust Localization and Perception in Contactless Delivery Vehicles: A Hybrid Approach Combining Extended Kalman Filtering and Transformer Networks Zuo, W. 2026 IEEE International Conference on Power, Electronics and Green Energy, ICPEGE 2026, 2026 https://www.scopus.com/pages/publications/105037462121?origin=scopusAI

38. Real-time location discovery with IMU, GPS, and laser range finder Hameed, M.A., Sulttan, M.Q., Wannas, A.A. AIP Conference Proceedings, 2025 https://www.scopus.com/pages/publications/105023303854?origin=scopusAI

39. Sustainable IoT-Enabled Parking Management: A Multiagent Simulation Framework for Smart Urban Mobility Mutambik, I. Sustainability (Switzerland), 2025 https://www.scopus.com/pages/publications/105011632573?origin=scopusAI

40. An Internet of Things Based Intercity Bus Management System for Smart City Taufik, M., Hudiono, H., Rakhmania, A.E., (...), Sari, A.S. International Journal of Computing and Digital Systems, 2021 https://www.scopus.com/pages/publications/85122736657?origin=scopusAI

41. Enhancing Train Transportation in Sri Lanka: A Smart IOT Based Multi-Subsystem Approach Using MQTT Balasingham, D., Samarathunga, S., Bandara, A., (...), Wijayaraja, J.L. IEEE Region 10 Annual International Conference, Proceedings/TENCON, 2024 https://www.scopus.com/pages/publications/105000224662?origin=scopusAI

42. Smart and integrated ticketing in the UK: Piecing together the jigsaw Turner, M., Wilson, R. Computer Law and Security Review, 2010 https://www.scopus.com/pages/publications/77949551991?origin=scopusAI

43. DESIGN OF A SECURE DATA TRANSMISSION SYSTEM BETWEEN TWO COMPUTERS USING COLUMNAR TRANSPOSITION AND AES ENCRYPTION OVER THE TCP/IP PROTOCOL Linh, V.T., Giang,

L.N. ARPN Journal of Engineering and Applied Sciences, 2025
https://www.scopus.com/pages/publications/105018382108?origin=scopusAI

44. Enhancing Secure Data Transfer Through Blockchain, Steganography, and Advanced Cryptographic Algorithms (RC5 and AES) Sultana, A.N., Suseela, G. 2025 International Conference on Networks and Cryptology, NETCRYPT 2025, 2025 https://www.scopus.com/pages/publications/105015390965?origin=scopusAI

45. Secure data dissemination for intelligent transportation systems Sun, L., Du, Q. SpringerBriefs in Computer Science, 2017 https://www.scopus.com/pages/publications/85044932564?origin=scopusAI

46. CHECK BUS - Bus Monitoring System for MTC Buses Valarmathi, V., Prasanth Raj, M., Srihari, V. 2023 Intelligent Computing and Control for Engineering and Business Systems, ICCEBS 2023, 2023
https://www.scopus.com/pages/publications/85189151253?origin=scopusAI

47. Accessible participatory design: Engaging and including visually impaired participants Andrews, C. Inclusive Designing: Joining Usability, Accessibility, and Inclusion, 2014
https://www.scopus.com/pages/publications/84925851691?origin=scopusAI

48. Designing with and for people living with visual impairments: audio-tactile mock-ups, audio diaries and participatory prototyping Metatla, O., Bryan-Kinns, N., Stockman, T., Martin, F. CoDesign, 2015
https://www.scopus.com/pages/publications/84925640469?origin=scopusAI

49. Introducing Diversity Personas in Gender-Responsive Mobility Planning Mattozzi, A., Princivalle, L., Diana, M., (...), Marsden, N. Lecture Notes in Computer Science, 2025
https://www.scopus.com/pages/publications/105007997448?origin=scopusAI

50. Persona Design Methodology for Work-Commute Travel Behaviour Using Latent Class Cluster Analysis Rasca, S.I., Markvica, K., Biesinger, B. Multimodal Transportation, 2023
https://www.scopus.com/pages/publications/85172665489?origin=scopusAI

51. Travel experience toolkit: Bus-specific tools for digital service design Hildén, E., Väänänen, K., Chistov, P. ACM International Conference Proceeding Series, 2018
https://www.scopus.com/pages/publications/85059979494?origin=scopusAI

52. AI-Supported Personas vs. Conventional Personas: A Comparative Study Based on the Views and Opinions of Designers Balci, E.C., Alppay, E.C. Applied Human Factors and Ergonomics International, 2025
https://www.scopus.com/pages/publications/105031255411?origin=scopusAI

53. Smart, public buses information system Megalingam, R.K., Raj, N., Soman, A.L., (...), Vijay, D. International Conference on Communication and Signal Processing, ICCSP 2014 - Proceedings, 2014
https://www.scopus.com/pages/publications/84916242830?origin=scopusAI

54. Bus Tracking System using Mobile GPS Technology Jayaraam, S.P., Jayakumar, D., Sandhiya, B., (...), Karthika, J. 7th International Conference on Inventive Computation Technologies, ICICT 2024, 2024
https://www.scopus.com/pages/publications/85196090686?origin=scopusAI

55. Impact of the Real-time Bus Information on Bus Riders, an Evidence from Guangzhou Liu, J., Mei, J. Journal of Harbin Institute of Technology (New Series), 2019 https://www.scopus.com/pages/publications/85076251588?origin=scopusAI

56. An experiment evaluating the impacts of real-time transit information on bus riders in Tampa, Florida Brakewood, C., Barbeau, S., Watkins, K. Transportation Research Part A: Policy and Practice, 2014
https://www.scopus.com/pages/publications/84907960912?origin=scopusAI

57. Ecotrips: Leveraging co-benefits and metaphorical metrics in a mobile app to promote walking and biking for short trips Park, H., Sanguinetti, A., Cortes, G.C. Lecture Notes in Computer Science (including subseries Lecture

Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2017
https://www.scopus.com/pages/publications/85025126759?origin=scopusAI

58. Emotion-Aware In-Car Feedback: A Comparative Study Mwaita, K.F., Bhaumik, R., Ahmed, A., (...), Haller, M. Multimodal Technologies and Interaction, 2024 https://www.scopus.com/pages/publications/85199528417?origin=scopusAI

59. Automatic identification of vehicles with faulty automatic vehicle location and control units in London buses' ibus system Robinson, S., Manela, M. Transportation Research Record, 2012
https://www.scopus.com/pages/publications/84869194259?origin=scopusAI

60. A gradient boosting decision tree based GPS signal reception classification algorithm Sun, R., Wang, G., Zhang, W., (...), Ochieng, W.Y. Applied Soft Computing Journal, 2020
https://www.scopus.com/pages/publications/85076541901?origin=scopusAI

61. GPS data processing for driving cycle development in Hanoi, Vietnam Nguyen, Y.-L.T., Bui, N.D., Nghiem, T.-D., Le, A.-T. Journal of Engineering Science and Technology, 2020
https://www.scopus.com/pages/publications/85084126653?origin=scopusAI

62. Urban bus arrival time prediction using linear regression and kalman filter—a comparison Ramkumar, N., Chaudhari, A. Advances in Intelligent Systems and Computing, 2019
https://www.scopus.com/pages/publications/85062281462?origin=scopusAI

63. Enabling bus transit service quality co-monitoring through smartphone-based platform Li, C., Zegras, P.C., Zhao, F., (...), Zhao, J. Transportation Research Record, 2017
https://www.scopus.com/pages/publications/85057972223?origin=scopusAI

64. An Adaptive Communication Model for Android Bus Tracking App Humaira, B.B., Chiniah, A. 2021 2nd Global Conference for Advancement in Technology, GCAT 2021, 2021
https://www.scopus.com/pages/publications/85119510092?origin=scopusAI

65. IoT-Based College Bus Tracking and Monitoring System Satyanarayana, S.V., Manasa, M., Vishwanth, B., (...), Sai, D.A. 2023 3rd International Conference on Artificial Intelligence and Signal Processing, AISP 2023, 2023
https://www.scopus.com/pages/publications/85163197423?origin=scopusAI

66. Edge-Based Facial Recognition System for Smart School Bus Monitoring Using IoT Salhi, D.E., Rawashdeh, M., Alnusair, A., Karime, A. 2026 International Conference on AI Innovations and Industry, ICAIII 2026 - Conference Proceedings, 2026 https://www.scopus.com/pages/publications/105041490912?origin=scopusAI

67. Real-Time Bus Tracking Using LoRa Technology Raghavi, Y., Chakravarthy, V.V.S.S.S., Sri, R.S., (...), Vivek, V. 4th Wireless, Antenna and Microwave Symposium, WAMS 2025, 2025
https://www.scopus.com/pages/publications/105018108888?origin=scopusAI

68. Inequality in User Experience: Can Mobile User Interfaces that Help Sighted Users Create Barriers for Visually Challenged People? Tanwar, S., Rao, P.V.M. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 2024
https://www.scopus.com/pages/publications/85200336964?origin=scopusAI

69. User-Centred Design in Educational Applications: A systematic literature review Al-Sa'di, A., McPhee, C.C.A. Proceedings - 2021 International Conference Engineering Technologies and Computer Science, EnT 2021, 2021
https://www.scopus.com/pages/publications/85124007186?origin=scopusAI

70. Tutorial on UCD Sprint: Inclusive Process for Concept Design Larusdottir, M.K., Roto, V., Lanzilotti, R., Visescu, I.D. ACM International Conference Proceeding Series, 2022
https://www.scopus.com/pages/publications/85139869101?origin=scopusAI

71. A comparative study on the role of models and prototypes in human-centered design versus design-driven innovation approaches Isa, S.S., Liem, A. Proceedings of the International Conference on Engineering Design, ICED, 2015 https://www.scopus.com/pages/publications/84979732915?origin=scopusAI